

SQL Code

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1  -- Databricks notebook source
2  -- DBTITLE 1,Introduction
3  -- MAGIC %md
4  -- MAGIC # Phase 1: Selection and Filteration (SQL)
5  -- MAGIC
6  -- MAGIC This notebook demonstrates fundamental SQL data operations:
7  -- MAGIC - **Selection**: Choosing specific columns using SELECT
8  -- MAGIC - **Filteration**: Applying WHERE conditions to filter rows
9  -- MAGIC
10 -- MAGIC ## Objectives:
11 -- MAGIC 1. Load and preview customer data
12 -- MAGIC 2. Select specific columns
13 -- MAGIC 3. Apply filtering conditions with WHERE
14 -- MAGIC 4. Combine multiple filters
15 -- MAGIC 5. Analyze filtered results
16
17 -- COMMAND -----
18
19 -- DBTITLE 1,Data Overview
20 -- Load and preview customer data
21 SELECT
22     COUNT(*) as total_records,
23     COUNT(DISTINCT customer_name) as unique_customers
24 FROM workspace.default.customer_report;
25
26 -- COMMAND -----
27
28 -- DBTITLE 1,Preview All Columns
29 -- Preview sample data (all columns)
30 SELECT *
31 FROM workspace.default.customer_report
32 LIMIT 10;
33
34 -- COMMAND -----
35
36 -- DBTITLE 1,Column Selection
37 -- SELECTION: Choose specific columns
38 SELECT
39     customer_name,
40     city,
41     segment,
42     total_spend,
43     order_count
44 FROM workspace.default.customer_report
45 LIMIT 10;
46
47 -- COMMAND -----
48
49 -- DBTITLE 1,Filter by Segment
50 -- FILTERATION: Filter by segment (Gold customers only)
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```
51 SELECT
52     customer_name,
53     city,
54     segment,
55     total_spend,
56     order_count
57 FROM workspace.default.customer_report
58 WHERE segment = 'Gold'
59 LIMIT 10;
60
61 -- COMMAND -----
62
63 -- DBTITLE 1,Gold Customer Count
64 -- Count Gold segment customers
65 SELECT
66     COUNT(*) as gold_customers
67 FROM workspace.default.customer_report
68 WHERE segment = 'Gold';
69
70 -- COMMAND -----
71
72 -- DBTITLE 1,Filter by Spend
73 -- FILTRATION: High-value customers (spend > $5M)
74 SELECT
75     customer_name,
76     city,
77     segment,
78     total_spend,
79     order_count
80 FROM workspace.default.customer_report
81 WHERE total_spend > 5000000
82 ORDER BY total_spend DESC
83 LIMIT 15;
84
85 -- COMMAND -----
86
87 -- DBTITLE 1,High-Value Count
88 -- Count high-value customers
89 SELECT
90     COUNT(*) as high_value_customers
91 FROM workspace.default.customer_report
92 WHERE total_spend > 5000000;
93
94 -- COMMAND -----
95
96 -- DBTITLE 1,Combined Filters
97 -- COMBINED FILTERS: Gold segment AND high-value
98 SELECT
99     customer_name,
100     city,
101     total_spend,
102     order_count
103 FROM workspace.default.customer_report
104 WHERE segment = 'Gold'
```

```
105     AND total_spend > 5000000
106 ORDER BY total_spend DESC
107 LIMIT 10;
108
109 -- COMMAND -----
110
111 -- DBTITLE 1,City Distribution
112 -- City distribution for Gold high-value customers
113 SELECT
114     city,
115     COUNT(*) as customer_count
116 FROM workspace.default.customer_report
117 WHERE segment = 'Gold'
118     AND total_spend > 5000000
119 GROUP BY city
120 ORDER BY customer_count DESC
121 LIMIT 10;
122
123 -- COMMAND -----
124
125 -- DBTITLE 1,Filter by Activity
126 -- FILTERATION: Active customers (order_count >= 30)
127 SELECT
128     customer_name,
129     city,
130     segment,
131     total_spend,
132     order_count
133 FROM workspace.default.customer_report
134 WHERE order_count >= 30
135 ORDER BY total_spend DESC
136 LIMIT 10;
137
138 -- COMMAND -----
139
140 -- DBTITLE 1,Active Customer Stats
141 -- Segment breakdown for active customers
142 SELECT
143     segment,
144     COUNT(*) as customer_count,
145     SUM(total_spend) as total_revenue,
146     AVG(total_spend) as avg_spend
147 FROM workspace.default.customer_report
148 WHERE order_count >= 30
149 GROUP BY segment;
150
151 -- COMMAND -----
152
153 -- DBTITLE 1,Summary Statistics
154 -- PHASE 1 SUMMARY: All filter results
155 SELECT
156     'Total Customers' as metric,
157     COUNT(*) as count
158 FROM workspace.default.customer_report
```

```
159
160 UNION ALL
161
162 SELECT
163     'Gold Customers' as metric,
164     COUNT(*) as count
165 FROM workspace.default.customer_report
166 WHERE segment = 'Gold'
167
168 UNION ALL
169
170 SELECT
171     'High-Value (>$5M)' as metric,
172     COUNT(*) as count
173 FROM workspace.default.customer_report
174 WHERE total_spend > 5000000
175
176 UNION ALL
177
178 SELECT
179     'Gold + High-Value' as metric,
180     COUNT(*) as count
181 FROM workspace.default.customer_report
182 WHERE segment = 'Gold' AND total_spend > 5000000
183
184 UNION ALL
185
186 SELECT
187     'Active (>=30 orders)' as metric,
188     COUNT(*) as count
189 FROM workspace.default.customer_report
190 WHERE order_count >= 30
191
192 ORDER BY
193     CASE
194         WHEN metric = 'Total Customers' THEN 1
195         WHEN metric = 'Gold Customers' THEN 2
196         WHEN metric = 'High-Value (>$5M)' THEN 3
197         WHEN metric = 'Gold + High-Value' THEN 4
198         WHEN metric = 'Active (>=30 orders)' THEN 5
199     END;
200
201 -- COMMAND -----
202
203
```